

The Effectiveness of Generative AI in Creating Personalised Video and Audio Content: A Literature Review

Abstract

Generative Artificial Intelligence (GenAI) has rapidly changed the way digital content can be created, particularly in the areas of video, music, and speech generation. Recent advances in text-to-video, text-to-music, and speech synthesis models allow users to generate personalised digital content with less technical knowledge and reduced production time. However, the effectiveness of these technologies cannot be evaluated only through their ability to produce realistic outputs. Issues relating to content quality, personalisation, production efficiency, authenticity, copyright, privacy, misinformation, and human creativity also need to be considered. This literature review examines recent research on generative AI for video and audio content creation, focusing on three main dimensions: content quality and personalisation, production efficiency, and ethical challenges. The reviewed literature indicates that GenAI can significantly improve accessibility, scalability, and the speed of digital content production. At the same time, limitations remain in temporal consistency, semantic accuracy, controllability, authenticity, and ethical governance. The literature also shows an imbalance between technical development and research into potential negative impacts. Therefore, the effectiveness of generative AI should be understood as a combination of technical performance, production benefits, human evaluation, and responsible human oversight. Further research is required to establish reliable evaluation frameworks and practical ethical guidelines for personalised AI-generated media.

Table of Contents

1. Introduction and Background
2. Purpose and Aims of the Literature Review
3. Literature Review Process
4. Main Body of the Report
 - 4.1 Generative AI for Video and Audio Content Creation
 - 4.2 Content Quality and Personalisation
 - 4.3 Production Efficiency and Accessibility
 - 4.4 Ethical Challenges
 - 4.5 Research Gaps and Critical Analysis
5. Conclusion
6. References

1. Introduction and Background

Generative Artificial Intelligence has become an important technology for digital content creation. Unlike traditional artificial intelligence systems that mainly analyse or classify existing information, generative AI can produce new content such as text, images, music, speech, and video. Recent advances in deep learning and generative models have made it

increasingly possible for users to create digital media from relatively simple instructions. This development has significant implications for creative industries because content production has traditionally required specialised technical skills, expensive software, and substantial amounts of time.

Generative AI is particularly relevant to video and audio production. Text-to-video systems can transform textual descriptions into video sequences, while text-to-music and speech-generation systems can create audio content based on user instructions. Models such as Make-A-Video, Imagen Video, VideoPoet, MusicLM, MusicGen, Voicebox, and NaturalSpeech demonstrate the rapid development of these capabilities. These technologies provide opportunities for content creators to produce advertising material, educational resources, entertainment, social media content, personalised communication, and other forms of digital media.

The development of these technologies also creates a shift in the role of human creators. Instead of producing every component manually, creators can increasingly use AI systems to generate initial content, modify existing material, or automate repetitive tasks. This can potentially reduce production costs and increase productivity. However, the quality and reliability of AI-generated content remain important concerns. Generated video may contain temporal inconsistencies or unrealistic physical behaviour, while generated audio may raise concerns regarding voice identity, authenticity, and unauthorised imitation.

The ethical dimension is equally important. Generative audio research, for example, has already identified concerns involving fraud, deepfakes, and copyright infringement. Barnett's systematic literature review of 884 generative-audio papers found that although many papers discussed positive applications, fewer than 10% discussed potential negative impacts. This suggests that technological development may be progressing faster than systematic consideration of its social consequences.

Therefore, this literature review examines the effectiveness of generative AI in creating personalised video and audio content. Effectiveness is considered from three perspectives: content quality and personalisation, production efficiency, and ethical responsibility.

2. Purpose and Aims of the Literature Review

The purpose of this literature review is to critically examine recent academic research on the use of generative AI for personalised video and audio content creation.

The review focuses on three research questions:

RQ1: How does generative AI affect the quality, realism, creativity, and relevance of personalised video and audio content?

RQ2: How does generative AI improve the efficiency, accessibility, and scalability of digital content production?

RQ3: What ethical challenges are associated with AI-generated video and audio, particularly in relation to authenticity, copyright, privacy, misinformation, and human creativity?

The review aims not only to summarise previous studies but also to compare their findings, identify common trends, evaluate methodological limitations, and identify areas requiring further research.

3. Literature Review Process

The literature for this review was selected from academic publications, conference proceedings, journal articles, and reputable scholarly repositories. The search focused primarily on research published between 2020 and 2026, consistent with the assessment requirement to prioritise recent literature.

Search terms included combinations of the following keywords:

- "generative AI"
- "AI-generated video"
- "text-to-video generation"
- "generative audio"
- "AI music generation"
- "speech generation"
- "voice cloning"
- "personalised digital content"
- "content quality"
- "production efficiency"
- "deepfake"
- "AI ethics"
- "copyright"
- "synthetic media"

The literature was organised according to four major themes: technological development, content quality and personalisation, production efficiency, and ethical challenges. Priority was given to peer-reviewed journal articles and established conference proceedings. Technical papers describing major generative models were also considered because they provide evidence of the capabilities and limitations of current systems.

The selected literature was critically compared according to research objectives, methodologies, findings, strengths, and limitations. Rather than treating each study independently, findings were synthesised across themes to identify broader patterns in the literature.

4. Main Body of the Report

4.1 Generative AI for Video and Audio Content Creation

Recent research demonstrates rapid progress in generative video and audio technologies. In video generation, models such as Make-A-Video and Imagen Video demonstrate that text prompts can be used to create visual sequences. Later systems, including VideoPoet, expanded this capability by supporting multiple modalities, including text, images, video, and audio. This development is significant because digital content is increasingly multimodal rather than limited to a single medium.

Generative audio has developed in a similar direction. MusicLM demonstrates the ability to generate music from textual descriptions, while MusicGen provides greater control over generated musical content. Speech-generation systems such as Voicebox and NaturalSpeech 2 further demonstrate improvements in naturalness, multilingual generation, and speaker characteristics.

These developments suggest that GenAI can increasingly function as a general-purpose content creation tool. However, technical capability does not necessarily mean that generated content is equivalent to professionally produced content. The effectiveness of a system depends on factors including output quality, consistency, controllability, relevance to user requirements, and the amount of human intervention required.

An important trend in the literature is therefore the movement from simply asking whether AI can generate content to asking whether the generated content is useful, reliable, controllable, and appropriate for a particular purpose.

4.2 Content Quality and Personalisation

Content quality is one of the most important factors when evaluating generative AI. Improvements in generative models have resulted in increasingly realistic video and audio outputs. For example, research into text-to-video generation demonstrates progress in visual fidelity, semantic alignment, and temporal coherence. Similarly, research on generative audio demonstrates improvements in naturalness, musical quality, and speech generation.

However, quality is difficult to define using a single technical metric. AI-generated video can appear visually realistic while still containing incorrect object movements, physical inconsistencies, or differences between frames. This is particularly important because video contains both spatial and temporal information. A generated image may appear realistic as a single frame, while the same content may appear unrealistic when viewed as a sequence.

This limitation has encouraged researchers to investigate more comprehensive evaluation approaches. Recent survey research on AI-generated video evaluation argues that traditional video metrics may not fully capture semantic quality, human perception, and alignment with user intention. This indicates that human evaluation remains important when assessing whether generated content is genuinely effective.

Personalisation introduces another dimension of quality. A generated video or audio file may be technically high quality but still fail to meet the user's specific requirements. For example, personalised educational content needs to reflect the learner's level and context, while personalised advertising needs to match the target audience's preferences. Therefore, relevance and controllability are important components of effectiveness.

Generative audio models also demonstrate strong potential for personalisation. Text-to-music systems allow users to specify musical characteristics through natural-language prompts, while speech-generation models can generate voices in different languages and styles. These capabilities can reduce the technical expertise required to create customised audio.

Nevertheless, the literature suggests that personalisation can create a trade-off between flexibility and control. More complex prompts may not always result in outputs that accurately follow user intentions. Consequently, human creators may still need to review, modify, and regenerate content.

Overall, the literature indicates that generative AI has significantly improved the ability to create realistic and personalised media, but quality remains dependent on the specific application and evaluation criteria. Technical realism alone is not sufficient to establish effectiveness.

4.3 Production Efficiency and Accessibility

Production efficiency represents one of the strongest arguments for adopting generative AI. Traditional video and audio production can require multiple stages, including planning, recording, editing, sound design, visual effects, and post-production. These processes can require specialised personnel and significant financial resources.

Generative AI can automate parts of these workflows. Text-to-video systems can generate initial visual material directly from descriptions, while text-to-music and speech systems can produce audio without requiring a complete recording process. Voicebox, for example, demonstrates the potential for highly efficient speech generation, while multimodal models can integrate several content types within a single system.

This capability can make content creation more accessible to individuals and organisations that do not have extensive technical expertise. Small businesses, educators, independent creators, and social media users may be able to produce content that previously required professional production teams.

The literature also suggests that GenAI can increase scalability. A human creator can use an AI system to generate multiple variations of content for different audiences. This is particularly relevant to personalised digital marketing, where different users may receive content adapted to their interests, language, or demographic characteristics.

However, increased automation does not necessarily eliminate human labour. Instead, the nature of creative work may change. Human creators may increasingly focus on prompt design, creative direction, quality control, fact checking, editing, and ethical decision-making. Research on GenAI and the creative industries suggests that the technology may accelerate creative workflows while simultaneously creating concerns about the future role and value of human creative labour.

Therefore, production efficiency should not simply be measured by the time required to generate an initial output. A more comprehensive measure should include the time required for checking, correcting, editing, and approving the final product. If AI-generated content

requires extensive human correction, the actual efficiency gain may be smaller than the initial generation speed suggests.

4.4 Ethical Challenges

4.4.1 Authenticity and Deepfakes

One of the most significant ethical concerns surrounding AI-generated media is authenticity. As generated video and audio become increasingly realistic, audiences may find it difficult to determine whether content is genuine or artificially produced.

Deepfake technology demonstrates this problem clearly. Synthetic video and audio can be used to imitate real people and create statements or events that never occurred. This creates risks involving misinformation, fraud, reputational damage, and manipulation.

The problem is not limited to malicious applications. Even legitimate use of synthetic media may create uncertainty when audiences are unable to distinguish between authentic and generated content. This creates a need for effective disclosure and provenance mechanisms.

Research into deepfake creation and detection has therefore become an important area. However, detection systems may also face limitations because generative technologies continue to evolve. A detection approach that performs well against one generation of models may become less effective against future systems.

This creates an ongoing technological competition between generation and detection. As a result, content authenticity should not depend entirely on automated detection. Human judgement, platform policies, provenance systems, and transparent disclosure may also be necessary.

4.4.2 Copyright and Ownership

Copyright is another major challenge. Generative models are often trained using large quantities of existing content, raising questions about whether copyrighted material can be used in training and how ownership should be determined for generated outputs.

The problem becomes particularly complex for music and voice generation. AI-generated music may reproduce characteristics associated with existing artists, while voice-cloning systems may reproduce the identity of a real person without permission.

The literature indicates that existing copyright frameworks were largely developed around human authorship and traditional forms of content creation. GenAI challenges these assumptions because a generated work may involve the model developer, training-data creators, prompt writer, and human editor.

Therefore, responsibility and ownership cannot always be assigned to a single actor. Clearer legal and organisational frameworks are required to determine how AI-generated content should be attributed and used.

4.4.3 Privacy, Consent, and Voice Identity

Voice generation introduces a particularly important privacy concern. A person's voice can function as a recognisable part of their identity. Voice cloning can reproduce characteristics of an individual's speech even when that person is not involved in the production process.

This creates questions about consent. For example, should an individual be allowed to prevent an AI system from reproducing their voice? Should permission be required before using recordings to create a synthetic voice?

Research on voice cloning has highlighted both positive and negative applications. Voice synthesis can potentially support accessibility and preserve aspects of a person's voice, but the same technology can also enable impersonation and unauthorised commercial use.

Therefore, responsible voice generation requires mechanisms for consent, transparency, and control over personal identity.

4.4.4 Bias and Representation

Generative AI can also reproduce biases present in its training data. When users request personalised content, the system may generate content based on patterns learned from existing datasets. If these datasets contain cultural, gender, racial, or social biases, generated content may reproduce or amplify them.

Personalisation therefore creates a tension between relevance and fairness. A system designed to optimise content for a specific audience may unintentionally rely on stereotypes.

This issue is especially important in advertising, education, and public communication, where generated content can influence people's attitudes and behaviour. Ethical evaluation should therefore consider not only whether content is technically accurate but also whether it represents different groups fairly.

4.4.5 Human Creativity and Employment

Another concern is the impact of generative AI on creative professionals. Automation can reduce the need for certain production tasks, potentially changing employment patterns in areas such as music, video editing, voice acting, and visual production.

However, the literature does not necessarily indicate that human creativity will disappear. Instead, creative work may be transformed. Humans can continue to provide creative concepts, cultural understanding, emotional judgement, and ethical oversight while AI performs repetitive generation tasks.

The effectiveness of GenAI should therefore be considered in terms of human-AI collaboration rather than complete replacement. The strongest applications may be those in which AI increases human capability without removing human responsibility.

4.5 Research Gaps and Critical Analysis

The literature reveals several important research gaps.

First, there is an imbalance between research on technical performance and research on negative consequences. Barnett's review of 884 generative-audio papers found that less than 10% discussed negative impacts, despite identifying serious concerns involving fraud, deepfakes, and copyright infringement. This suggests that ethical evaluation has not developed at the same pace as technical innovation.

Second, there is no universally accepted definition of content quality for generative media. Technical metrics can measure aspects of visual or audio similarity, but they may not capture whether content is meaningful, appropriate, trustworthy, or useful to its intended audience. This is especially important for personalised content because effectiveness depends on the relationship between the generated output and the user's needs.

Third, much of the technical literature evaluates individual models under controlled conditions. Real-world content production is more complex. Professional workflows involve editing, fact checking, copyright considerations, audience expectations, and human approval. Future research should therefore evaluate GenAI in realistic production environments rather than relying only on model benchmarks.

Fourth, research should investigate long-term effects on creative professionals. Existing literature discusses concerns about human labour and creativity, but further empirical research is required to determine whether GenAI primarily replaces creative tasks, augments them, or creates new forms of creative employment.

Finally, research into ethical governance remains fragmented. Hagendorff's comprehensive scoping review identified 378 normative issues across 19 topic areas in generative AI ethics, demonstrating the breadth of the ethical landscape. However, identifying ethical concerns is different from developing practical mechanisms to address them. Future research should therefore focus on operational approaches such as content labelling, consent systems, provenance mechanisms, copyright management, and human review.

Overall, the literature suggests that the effectiveness of GenAI should be evaluated using a multidimensional framework. Technical quality, production efficiency, personalisation, user satisfaction, authenticity, ethical risk, and human oversight should all be considered.

5. Conclusion

Generative AI has demonstrated significant potential to transform personalised video and audio production. Recent models have improved the realism, controllability, and diversity of generated media while reducing the technical expertise and time required for content creation. These developments can provide substantial benefits for professional creators, businesses, educators, and individual users.

However, technological capability alone does not establish overall effectiveness. AI-generated media can still contain inconsistencies, fail to follow user intentions, and require human correction. More importantly, the increasing realism of synthetic media creates ethical challenges involving deepfakes, misinformation, copyright, privacy, consent, bias, and human creative labour.

The literature therefore supports a balanced view of generative AI. GenAI should not be considered simply as a replacement for human creators. Its greatest value may come from human-AI collaboration, where AI provides speed, scalability, and generation capabilities while humans provide creative direction, quality control, contextual understanding, and ethical judgement.

Future research should develop more comprehensive evaluation frameworks that combine technical measurements with human assessment and ethical criteria. Greater attention should also be given to real-world production environments, long-term effects on creative work, and practical governance mechanisms. Ultimately, responsible human oversight will remain essential if generative AI is to become an effective and sustainable technology for personalised digital content creation.